

Rean AI Instant Visual Tutor - Source Notes

This companion file summarizes assumptions, decision points, errors, and excluded future scope. The primary presentation artifact is `rean_ai_instant_visual_tutor_user_flow.pdf`.

Section	Content
Assumptions	• Firebase Authentication is the identity provider. • Flutter mobile owns the student experience. • Node.js backend owns sessions, persistence, audit logging, and app APIs. • FastAPI AI Tutor owns problem understanding, teaching strategy, verification routing, and AI-provider integration. • SymPy or equivalent verification is required before math is marked correct. • Khmer-first UX is assumed, with English support available.
Excluded Future Scope	• Parent dashboard • Teacher dashboard • Advanced pronunciation scoring • Full TRACE-CAG research architecture • Complex admin assignment workflows • Multi-level admin roles • Local LLM management • Advanced enterprise analytics • Offline AI tutoring
Decision Points	• Is the student authenticated? • Has onboarding been completed? • Which dashboard action was selected? • Is the problem clear and supported? • Was student work provided? • Which first intervention is safest? • Is the student response correct, partially correct, or incorrect? • Should support level increase? • Did mathematical verification pass? • Did the student request the full solution early? • Did the student complete guided steps? • Can the student apply the idea independently? • Is microphone permission granted? • Is the transcript accurate? • Is the admin authorized? • Is the reported response incorrect or unsafe? • Is retry available for the current error?

Section	Content
Error Cases	• No internet • Authentication failure • AI timeout • AI service unavailable • Invalid structured response • Failed verification • Unsupported problem • Image upload failure • Audio upload failure • Quiz loading failure • Save-progress failure • Expired session • Unauthorized admin request